

A chain-of-thought trace under Setting 1 (Multiple choice)

MedMCQA question

Cold agglutinin is

- ☐ A IgG
- ☒ B IgM keyed answer
- ☐ C IgA
- ☐ D IgD

Mistral Large 3

temperature 0 · reasoning emitted as visible output

Step-by-step reasoning:

- ❶ Cold agglutinins are antibodies that cause clumping (agglutination) of red blood cells at low temperatures, typically below normal body temperature.
- ❷ These antibodies are most active at temperatures around 4°C but can also react at higher temperatures (up to 30–37°C in some cases).
- ❸ Cold agglutinins are primarily associated with autoimmune hemolytic anemia and certain infections, such as *Mycoplasma pneumoniae* and Epstein-Barr virus.
- ❹ Immunoglobulin classes differ in structure and function. IgM is a pentameric antibody, making it highly efficient at agglutinating cells by binding multiple antigens simultaneously.
- ❺ IgG, IgA, and IgD are not typically associated with cold agglutination. IgG is more commonly involved in warm autoimmune hemolytic anemia.
- ❻ Therefore, cold agglutinins are predominantly of the IgM class.

Answer: (B) parsed as B · matches the key

The final line is the only part the multiple-choice parser reads; the numbered steps are the model's own formatting.